

First two (2) letters of *last* name:

11

Name (printed legibly): _____

Signature: _____

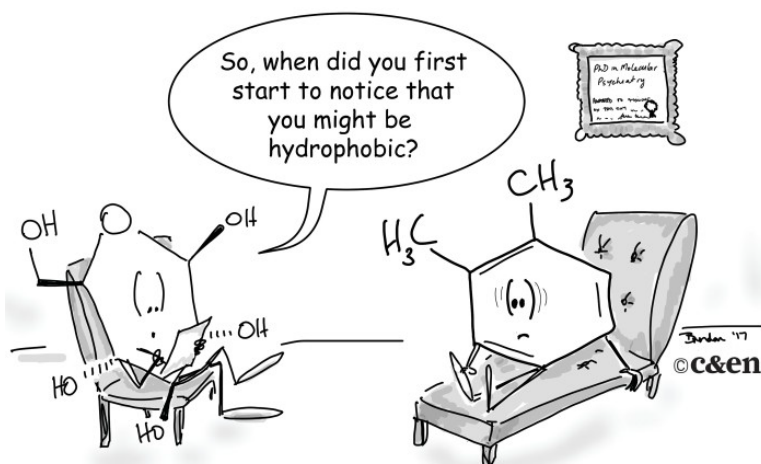
Student ID #:

as is continuing to write after time is called.

You may tear the periodic table/scratch paper off the back while waiting to begin.

We've covered a variety of chemistry where electrons work together beyond a single bonding interaction between atoms. Likewise I hope you have taken time to work together with classmates, peer-leaders, and me. For now, working together has come to a pause for this exam. Now as individuals – work carefully. Work completely. And work concisely using drawings!

Pete



page 2 (15)

page 3 _____ (14)

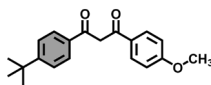
page 4-5 _____ (36)

page 6 (15)

page 7 (12)

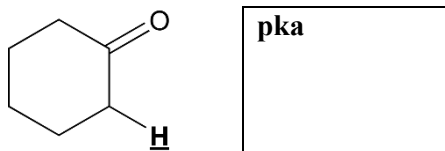
page 8 _____ (8)

TOTAL (100)



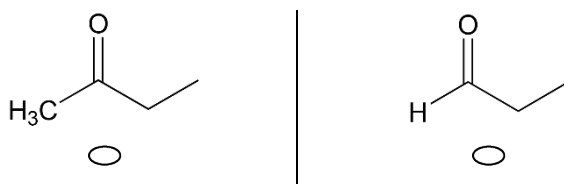
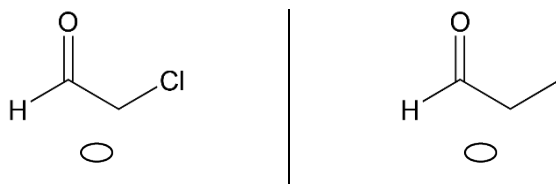
1. Draw a compound containing an aldehyde functional group and no others. (4)

2. Give the pKa of the bold, underlined hydrogen in the following compound. (4)

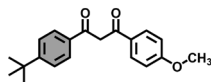


3. Consider the following pairs of carbonyl compounds.

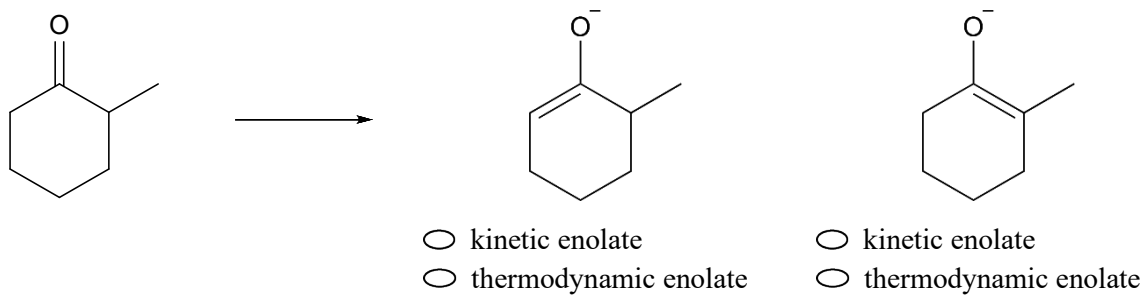
a) Choose which of the following has the most electrophilic sp^2 -hybridized carbon within each pair. Fill in the bubble. (4)



b) Rationalize your choice in the second pair. A correct, concise and relevant answer will earn maximum credit. (3)



4. Identify the following enolates as kinetically or thermodynamically favored and fill in the appropriate bubbles for each. (4)

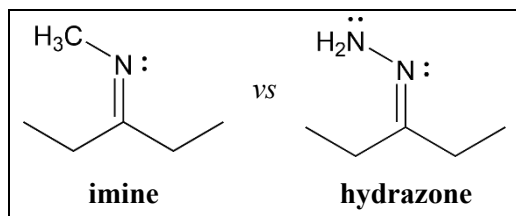


Which base and temperature would you use to favor formation of the kinetic enolate? (4)

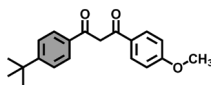
base: _____

temperature: _____

5. Hydrazones are not hydrolyzed as easily as imines and so they can be isolated. Explain why using a drawing (4 pts) and brief explanation (or label relevant parts of your drawing) (2 pts). (6)



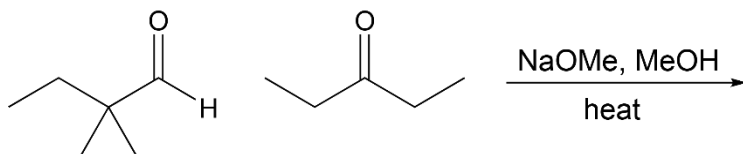
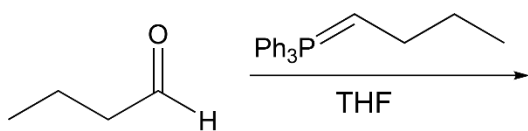
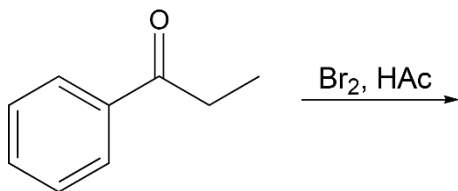
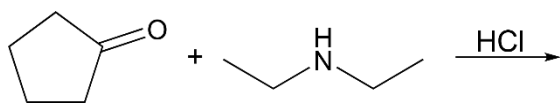
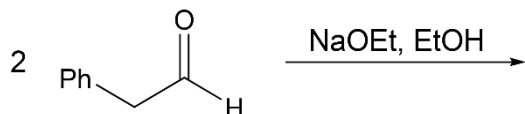
your initials
or signature

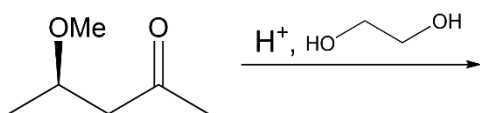
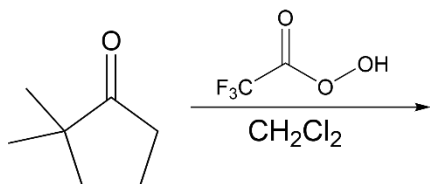
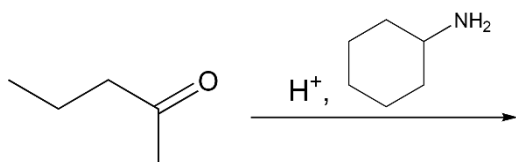
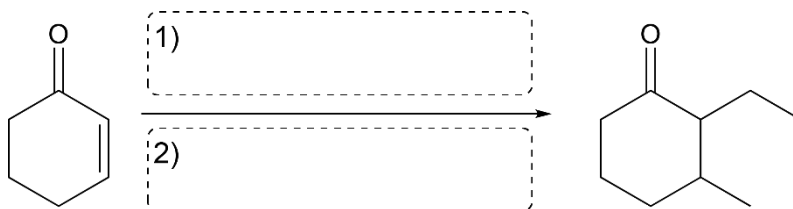
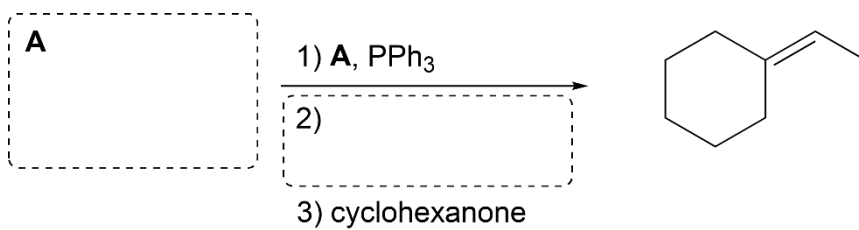
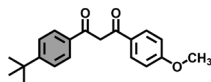


6. Complete the following reactions. In dashed boxes over arrows of reactions where products are shown, fill in the necessary conditions. Give the major product(s) of the other reactions. Be sure to show or otherwise indicate relevant stereochemical outcomes where appropriate. If no reaction takes place, write “no reaction” or “NR”.

Write out all reagents. For example, if the Bell Reagent was NaOEt, please write NaOEt or NaOCH₂CH₃ – not “Bell Reagent.” (36)

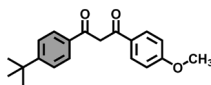
There are 10 reactions. **Complete nine. Write OMIT** in the answer space of the one you do not want graded. You can lose four (4) points if you do not write OMIT.





Did you remember to write OMIT for one reaction (in the answer space for that reaction)?

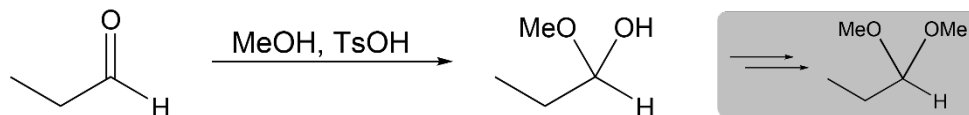
your initials
or signature



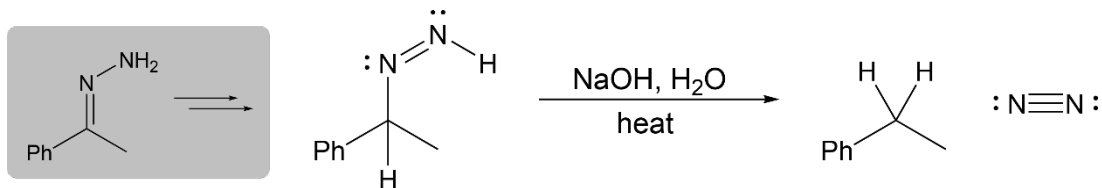
7. Complete the following mechanisms. You only need to give the mechanism for the unshaded reaction. The shaded boxes show the reactants or products for the full reaction.

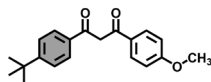
Write out the complete, step-by-step mechanism using curved arrows for the following reactions. Be certain that there is no ambiguity where arrows begin or end and that all bonds formed or broken are explicitly shown.

- a) (8)

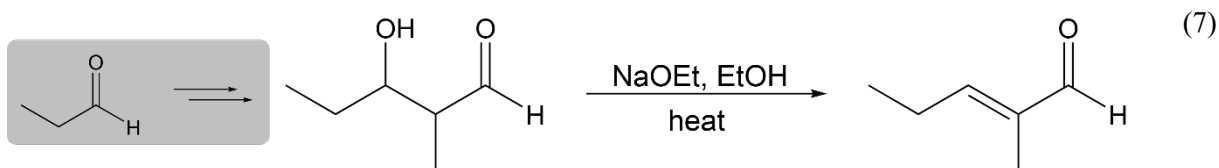


- b) (7)



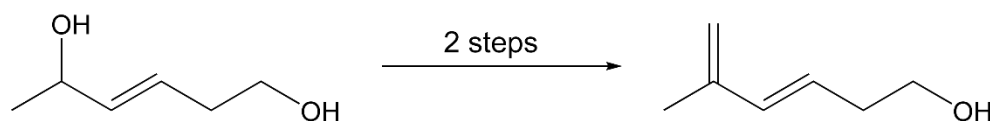


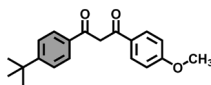
c)



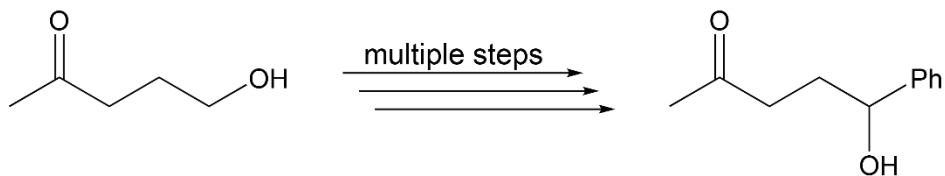
8. Complete the following synthesis using any other reagents needed. The best route allows completion in two steps. Partial credit will still be earned for other workable steps. Show each step and the product of each.

(5)





9. Propose a reasonable synthetic sequence for the following. Just show each individual step and the product of each. You should not draw the mechanisms (use scratch paper to draw those as needed, but just give each reaction here. (8)



1 IA 1A	Periodic Table of the Elements																18 VIIIA 8A
1 H Hydrogen 1.008	2 IIA 2A											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3 Li Lithium 6.941	4 Be Beryllium 9.012											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
11 Na Sodium 22.990	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B						
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [208.982]	85 At Astatine 209.987	86 Rn Radon 222.018
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [289]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]